

RISK REGISTER FOR 3rd YEAR PROJECT

|                |                                   |                  |                                         |
|----------------|-----------------------------------|------------------|-----------------------------------------|
| Project Title: | Bit-Serial RISC-V Array Processor | Submission Date: | April 29th 2026 (final updated version) |
| Student Name:  | Chi Tat Ip                        |                  |                                         |

| Project Risk                                                    | Severity |   |   | Potential |   |   | Score(Severity x Potential) | Mitigation Measures & Realised Outcome                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|----------|---|---|-----------|---|---|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                 | L        | M | H | L         | M | H | L=1, M=2, H=3               |                                                                                                                                                                                                                                                                                                           |
| Delays in FPGA delivery or hardware procurement                 |          | ✓ |   |           | ✓ |   | 4                           | Order parts early, identify alternative suppliers.<br><b>Outcome:</b> FPGA board (PYNQ-Z2) procured in W1 of Semester 1 with no procurement delay, risk did not materialise.                                                                                                                              |
| Bugs in HDL code                                                |          | ✓ |   |           |   | ✓ | 6                           | Construct more advanced testbenches, study verification techniques.<br><b>Outcome:</b> Bit-serial timing and writeback bugs surfaced during Sobel integration. Self-checking testbenches and waveform inspection in Vivado were used to localise faults. The bug was fully resolved before FPGA bring-up. |
| Integration issues between multiple bit-serial cores            |          |   | ✓ |           |   | ✓ | 9                           | Start with small-scale arrays (2×2 or 4×4) before scaling.<br><b>Outcome:</b> Mitigation effective. Integration was verified on a 2×2 array first, then scaled to 32×32 incrementally. Each scaling step exposed only minor address-decoder issues, all fixed without redesign.                           |
| Limited documentation or reference for bit-serial RISC-V arrays | ✓        |   |   |           |   | ✓ | 3                           | Review existing research, seek guidance from supervisor.<br><b>Outcome:</b> Addressed through extended literature review and weekly supervisor meetings, this directly informed the ISA simplification and NEWS interconnect design.                                                                      |
| Insufficient time for project milestones                        | ✓        |   |   |           | ✓ |   | 2                           | Create a detailed project plan with milestones and weekly progress tracking.<br><b>Outcome:</b> Front-loaded schedule absorbed short-notice job interview commitments.                                                                                                                                    |
| Hardware failures (FPGA, connectors, cables)                    |          | ✓ |   | ✓         |   |   | 2                           | Handle hardware carefully and maintain spares.<br><b>Outcome:</b> FPGA board and JTAG cable remained operational throughout.                                                                                                                                                                              |
| Communication / synchronisation bottlenecks between cores       |          | ✓ |   |           |   | ✓ | 6                           | Design scalable interconnect, test small arrays first.<br><b>Outcome:</b> NEWS-only interconnect kept synchronisation purely local. The OR-reduced data-out path caused routing congestion at 32×32, mitigated by adding a pipeline register on the SIPO collection net to close timing.                  |
| Loss of project data (code, simulations, results)               |          |   | ✓ |           | ✓ |   | 6                           | Use version control (Git) with regular pushes to a remote repository.<br><b>Outcome:</b> Risk realised when the development laptop failed. All RTL, testbenches and the report draft were recovered from the remote Git repository within one day.                                                        |
| Schedule clash with job interviews and assessments              |          | ✓ |   |           |   | ✓ | 6                           | Front-load deliverables to earlier weeks and maintain slack in later weeks.<br><b>Outcome:</b> Interviews were absorbed without milestone slippage because critical-path RTL had been pulled forward. Strategy retained for S2.                                                                           |
| Lit-review and architecture phase overrun (NEW – S1)            |          | ✓ |   |           | ✓ |   | 4                           | Allow design and prototyping to run in parallel rather than sequentially once core ISA decisions are stable.<br><b>Outcome:</b> Lit review extended by one week, recovered by overlapping with bit-serial ALU and register-file prototyping. No downstream impact.                                        |